Internship Opportunity – Classification of Intrusion Events Using DAS

Background

Distributed Acoustic Sensing (DAS) turns an optical fibre into a continuous array of acoustic sensors capable of measuring vibrations over several kilometres. Its operation relies on Rayleigh backscattering: when a light pulse travels through the fibre, a small fraction of the light is scattered back towards the source due to microscopic irregularities in the glass. The phase and amplitude of the backscattered light change when the fibre is strained, making it possible to detect and locate vibrational events.
The SURFO team uses this technology to monitor railway infrastructure. Applications include intrusion detection near railway tracks, rockfall identification and train traffic monitoring. For intrusion detection, identifying a disturbance is not sufficient: the event must also be classified (pedestrian, digging, fence climbing, vehicle, animal, passing train, etc.) in order to prioritise alerts, reduce false alarms and trigger an appropriate response. This internship focuses on event classification, downstream of the detection or segmentation of regions of interest.

Challenge

Genuine intrusions (people or animals) must be distinguished from normal sources of disturbance (trains, wind, maintenance work). Published methods report recognition rates above 97 % on small datasets, but performance falls to 60–88 % at real-world sites [1, 3, 4, 6–8] due to noise, class imbalance and a limited number of annotations.

Internship objective

Automatically classify intrusion events and assess whether analysing their temporal sequence improves decision-making.

Internship tasks

The work will focus on the following topics:

- **Literature and data review**: study the SURFO data and existing event-classification methods.
- **Annotated dataset**: define event classes, annotation guidelines and training/test splits by date and site.
- **Event classification**: develop a baseline method, then evaluate neural networks (CNN, CNN-LSTM and Transformer) using different data representations.
- **Sequence analysis**: investigate sequences of events over time (HMM, GRU/LSTM and trajectory tracking) to assess whether an intrusion is likely.
- **Evaluation and documentation**: assess robustness to noise, changes of site and unknown events, measure false alarms, and produce reproducible Python code.

Priority will be given to dataset preparation and event classification. Sequence analysis may be developed further depending on progress.

Expected outcomes

- A reference dataset, an event taxonomy and a documented evaluation protocol;
- An event-classification prototype with confidence scores and rejection of unknown classes, potentially extended with temporal analysis;
- A comparison of data representations and models by class and site, together with a robustness assessment and recommendations.

Candidate profile

Education

Final-year engineering student, second-year Master's student or student on a gap year, with a background in signal processing, artificial intelligence or applied mathematics.

Technical skills

- Strong foundations in mathematics, signal processing and/or *machine learning*;
- Proficiency in Python (NumPy, SciPy and PyTorch are appreciated);
- Ability to analyse experimental data.

Personal qualities

- Scientific rigour, curiosity and independence;
- Teamwork and good communication skills.

Practical information

- Duration: 5 to 6 months
- Location: La Plaine-Saint-Denis, France
- Start date: February–March 2027
- Supervision: BEN AMARA Imen, RIHANI Mahran
- Allowance: in accordance with applicable regulations
- Benefits: SNCF travel benefits.

How to apply

Please send your CV and cover letter to all the following addresses:

- `ext.mahran.rihani@reseau.sncf.fr`
- `gabriel.papaiz@reseau.sncf.fr`
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References

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